

What Does Occlusion Cost? Quantifying and Removing a Distinct Failure Mode in Autonomous Industrial Bin Picking

Donato Caramia¹, Florian T. Pokorny², Giuseppe Triggiani³, Denis Ruffino³, David Naso¹, Paolo Roberto Massenio¹

Abstract—Robot learning is being asked to meet industrial standards of reliability and throughput, yet failures in unstructured bin picking are still reported as a single aggregate number. We argue that they should not be: occlusion-induced failure, in which a score-driven grasp planner targets a physically inaccessible surface, is a *separable* failure mode with a measurable cost, and unlike sensor noise or novel-object brittleness it can be substantially mitigated with a lightweight perception module rather than a hardware upgrade. We quantify this cost over more than 6,000 real grasp attempts on a suction-based cell. Adding ViTA-Seg, a single-pass ViT amodal segmentation model, and an occlusion-understanding layer that orders picks by occluder/free/occluded hierarchy raises success rate by 6.3 and 8.6 percentage points under single- and multi-pick policies, increases grasp attempts by over 22%, and cuts recovery events by a third. We complement benchmark accuracy with per-frame operational metrics that capture how perception quality translates into picking throughput.

Index Terms—Amodal segmentation, bin picking, occlusion-aware manipulation, suction grasping, vision transformers

I. MOTIVATION: WHICH FAILURES CAN PERCEPTION ACTUALLY FIX?

Recent advances in Vision–Language–Action models and World Models [1] have enabled increasingly general robotic behaviors across tasks and embodiments. The bar for deployment, however, is not benchmark accuracy: it is sustained reliability, throughput, and the number of human interventions per shift. Under that bar, a perceptual problem that is easy to overlook in static evaluation becomes dominant. In dense clutter, objects partially or fully hide one another, so the visible portion of an object is an incomplete representation of its physical extent. Policies conditioned on visible RGB or RGB-D observations therefore act on an impoverished scene model, and occlusion stops being an edge case and becomes the standing scene condition [2].

Failures in unstructured bin picking are usually reported as a single success-rate deficit, but they arise from at least three distinct causes: (i) hardware and sensing noise, (ii) limited model robustness on novel objects, and (iii) occlusion-induced errors, in which the planner selects a surface that is scored

highly but is physically blocked or inaccessible. The distinction matters economically. Causes (i) and (ii) are addressed with better sensors, more data, or more capable end-effectors, all of which result in higher production costs and time for the cell. Cause (iii) is a representational deficiency: the planner is not wrong about the surface, it is blind to what lies on top of it. If that blindness can be removed with a small, single-pass perception module running inside the existing pipeline, it is by far the cheapest reliability improvement available to an integrator.

Current suction planners make exactly this mistake. Point-cloud [3] and RGB-D [4]–[6] methods are purely score-driven: they rate grasp quality on visible surfaces without modelling accessibility, and so favour large, prominent regions regardless of whether those regions can be reached. Amodal segmentation [7]–[9] could supply the missing representation, but is evaluated almost exclusively on static benchmarks [10], [11]. To our knowledge no prior work couples the two inside a running pipeline to quantify occlusion-induced failure or price its correction.

This extended abstract reports such a measurement. We ask two deployment questions:

- 1) *How much success rate and throughput does a state-of-the-art suction planner lose to occlusion alone?*
- 2) *What operational benefit does explicit amodal reasoning return when it is placed inside a complete, continuously running picking cell?*

Our contributions are:

- An occlusion-aware picking pipeline pairing ViTA-Seg, a single-pass ViT amodal model with negligible inference overhead, with a layer that labels objects occluder/free/occluded and orders picks accordingly (Sec. II);
- An evaluation protocol built for deployment rather than benchmarking, with 24h of operation, an instrumented recovery ladder, and two per-frame throughput metrics (Sec. III);
- A quantification of the occlusion-induced failure mode over 6,000+ real grasps, showing that perception *quality* and perception *cost* both convert directly into throughput (Sec. IV);
- An assessment of what still separates this pipeline from deployment (Sec. V).

¹Department of Electrical and Information Engineering, Polytechnic University of Bari, Bari, Italy. {d.caramia2@phd., david.naso@, paoloroberto.massenio@}poliba.it

²Division of Robotics, Perception and Learning (RPL), KTH Royal Institute of Technology, 11428 Stockholm, Sweden. fpokorny@kth.se

³R&D Department, AROL S.p.A., Canelli, Italy. {giuseppe.triggiani, denis.ruffino}@arol.com

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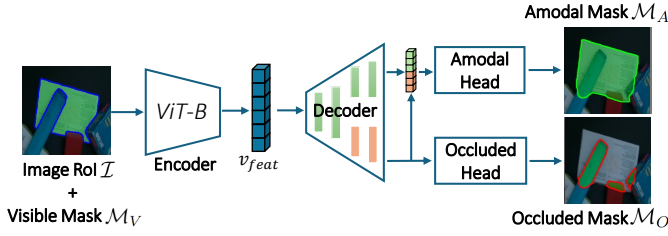


Fig. 1. ViTA-Seg architecture. A visible mask from SAM 2 and a 20%-padded RoI are mapped in a single forward pass to an amodal mask $\mathcal{M}_A$ and an occluded mask $\mathcal{M}_O$; pairwise overlap between $\mathcal{M}_O$ and neighbouring visible masks yields the occluder/free/occluded labelling used to order picks.

TABLE I
AMODAL ($mIoU_A$) AND OCCLUDED ($mIoU_O$) IOU ACROSS THREE BENCHMARKS.

Model	Par. (M)	KINS		COCOA		D2SA	
		$mIoU_A$	$mIoU_O$	$mIoU_A$	$mIoU_O$	$mIoU_A$	$mIoU_O$
C2F-Seg	193.50	87.89	57.30	87.15	36.69	90.58	38.92
ViTA-Seg	103.96	91.12	63.48	93.70	49.88	95.93	60.49

II. OCCLUSION-AWARE PICKING PIPELINE

A. Amodal perception module

ViTA-Seg (Fig. 1) takes two inputs: a 224×224 RoI $\mathcal{I}$ cropped around the visible-mask bounding box and padded by 20%, and the visible mask $\mathcal{M}_V$ itself, obtained class-agnostically from an upstream SAM 2 detector [12]. The two are concatenated into a 4-channel input to a ViT-B encoder pre-trained with MAE [13]; the extra input channel is initialised as the mean of the RGB projection weights. A convolutional decoder upsamples the $768 \times 14 \times 14$ token grid to 224×224 Amodal mask ($\mathcal{M}_A$) and Occluded mask ($\mathcal{M}_O$). A shared trunk splits into an amodal and an occluded branch, and the occluded branch’s features are concatenated into the amodal branch at the final stage, so the amodal prediction is informed by explicit hidden-region cues. Both heads are supervised with BCE plus Dice loss, combined as $\mathcal{L} = \mathcal{L}_A + \lambda_O \mathcal{L}_O$ with $\lambda_O = 0.25$.

What matters industrially is what the architecture omits: ViTA-Seg is strictly single-pass, with no iterative decoding, diffusion refinement, or shape-prior retrieval. It runs in 8.96 ms per instance against 113.77 ms for the closest class-agnostic competitor C2F-Seg [9], a $12.7\times$ speed-up on the same hardware (NVIDIA RTX 4090, i9-14900KF), at 103.96M parameters, roughly half of C2F-Seg’s. It thus co-resides on one GPU with SAM 2 and the grasp planner at negligible overhead. Sec. IV-C shows this is not cosmetic: it converts into additional grasp attempts per hour.

B. Does the perception layer hold up where it has to?

A perception module is only a reliability layer if it degrades gracefully under the conditions the cell actually produces, so we report two checks rather than a leaderboard. Trained and evaluated independently on three visually unrelated benchmarks (COCOA [10], KINS [11], D2SA [7]), ViTA-Seg leads C2F-Seg on amodal and occluded IoU on all three, by 3.2–6.5 and 6.2–21.6 points respectively, at half the parameters;

the consistency across domains speaks to the novel-object robustness of failure cause (ii) in Sec. I. The margin is largest precisely on occluded IoU, the hidden-region quantity that pick ordering depends on, reaching +13.2 points on COCOA and +21.6 on D2SA, so the model is most accurate exactly on the measurement the score-driven planner is blind to.

C. Occlusion understanding

Given N detected objects, an object i is a candidate occluded if $|\mathcal{M}_{O,i}| > \tau_{occ}$. For each candidate we measure the overlap between $\mathcal{M}_{O,i}$ and the $\mathcal{M}_{V,j}$ of every other object $j \neq i$; if it exceeds τ_{ov} we set $C_{j,i} = 1$ in a binary occlusion matrix $C \in \{0, 1\}^{N \times N}$. Objects with $\sum_j C_{j,i} > 0$ are occluded (O_d); objects occluding at least one other are occluders (O_r); the rest are free (F). We use $\tau_{occ} = 200$ and $\tau_{ov} = 40$ pixels. The output is a scene representation, which is what makes multi-pick sequencing possible.

D. Grasping policies

Both policies use CoAS-Net [5] suction scores and are run with and without ViTA-Seg, so the perception layer is the only variable.

Single-Shot Single-Pick (SSSP). One RGB-D acquisition per grasp. The baseline picks the highest-scoring visible instance; the occlusion-aware variant restricts the choice to occluders O_r , i.e. to objects nothing is resting on, and picks the highest-scoring one among them.

Single-Shot Multi-Pick (SSMP). One acquisition yields an ordered sequence of picks [14]. The baseline sorts all instances by score. The occlusion-aware variant groups them as O_r , F , O_d , sorts by score within each group, and concatenates. SSSP isolates the effect of occlusion-aware *target selection*; SSMP tests whether the full occlusion hierarchy buys *throughput*.

III. AN EVALUATION PROTOCOL FOR DEPLOYMENT, NOT FOR BENCHMARKS

The cell (Fig. 2) is a UFACTORY xArm 6-DoF manipulator with the manufacturer’s vacuum gripper and a wrist-mounted Intel RealSense D405 at approximately 35 cm. Two adjacent bins hold 24 objects of varied geometry, texture, and size, arranged to maximise inter-object occlusion.

Each configuration, defined by the pairing of a grasping policy with a perception module, runs for a *5-hour fully automated session*: SSSP and SSMP with and without ViTA-Seg, plus 2-hour session of SSMP with C2F-Seg. Objects are transferred bin to bin and released at random over the target bin, so every cycle regenerates a fresh cluttered arrangement rather than replaying a fixed scene.

Because the system must not stall, recovery is explicit and instrumented. After a failed grasp the robot attempts the next-highest-priority object in the current scene; if that also fails, the scene is re-acquired and perception restarts; after two further consecutive failures the robot switches bins. A separate safety routine returns the arm to a home configuration on joint singularities or unexpected collisions. Recovery events

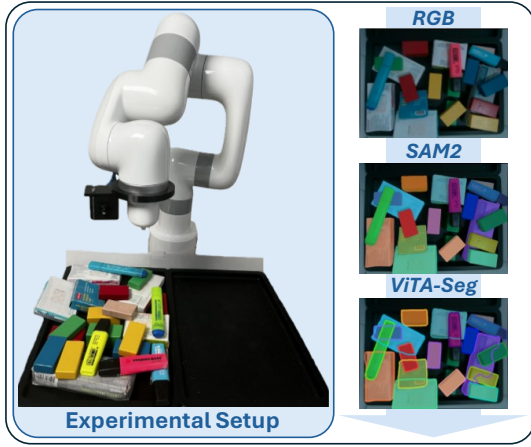


Fig. 2. The cell: UFACTORY xArm with vacuum gripper, wrist-mounted RealSense D405 at ~ 35 cm, two bins, and the 24-object set chosen to maximise inter-object occlusion.

are logged, and we treat their count as a proxy for operator-intervention load, which is what an integrator is actually buying down.

Success rate alone cannot express the value of multi-pick operation, so we report two per-frame metrics.

$$\text{SPF} = \frac{\text{Success Grasps}}{\text{Total Frames}} \quad \text{OPF} = \frac{\text{Success Grasps}}{\text{Total Frames}} \quad (1)$$

both measure how much work is extracted from a single perception step. A policy with identical success rate but higher SPF re-perceives less often, and in a cell where perception sits on the critical path this is a direct cycle-time saving. We propose SPF and OPF as reporting standards for multi-pick policies.

IV. WHAT OCCLUSION COSTS, AND WHAT IT COSTS TO FIX

A. Single-pick: the cost of occlusion in isolation

Under SSMP (Table II), occlusion-aware target selection raises success rate by 6.3 percentage points. The mechanism is visible in the raw counts: the number of grasp *attempts* is essentially unchanged (1050 vs. 1058), but 61 more of them succeed. A score-driven planner favours large, prominently visible surfaces irrespective of accessibility and collides with whatever rests on them; restricting selection to occluders removes that class of collision. Recovery events fall from 38 to 26, a 32% reduction in the events that would require attention on a real line. This is the answer to Question 1: with an occlusion-unaware planner, roughly one in sixteen attempts fails for a reason that is neither hardware nor generalisation, but representation.

B. Multi-pick: throughput, not just accuracy

SSMP shows the larger effect. Success rate rises 8.6 points, and within the same 5-hour window grasp attempts increase by 22.8% and successful grasps by 39.3%. Crucially, the number of scene acquisitions *falls* (296 vs. 312) while output rises, so

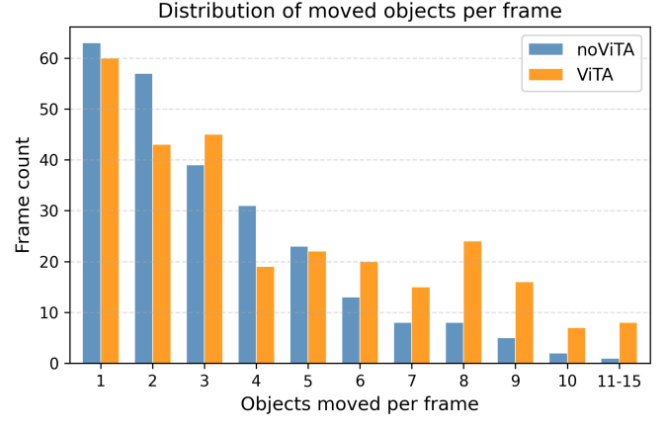


Fig. 3. Objects moved per frame under SSMP. Occlusion-aware sequencing shifts mass towards high-throughput frames rather than raising the mean uniformly.

SPF improves from 2.64 to 3.88 and OPF from 2.59 to 4.05: respecting the occlusion hierarchy lets the robot extract more picks per perception step before the scene model goes stale. Fig. 3 shows this is a distributional shift, not an averaging artefact: occlusion-aware sequencing needs fewer frames to clear five or fewer objects and holds a systematic advantage for scenes of six or more, up to the observed maximum of 15 objects from a single frame. Recovery events drop from 100 to 67. This answers Question 2: the benefit is not only fewer failed grasps but fewer perception cycles per unit of work.

C. Perception quality and perception cost both convert to throughput

Table III holds the pipeline fixed and swaps only the amodal module, over matched 2-hour sessions. Two results stand out. First, both amodal pipelines beat the occlusion-unaware baseline, so the *planning paradigm* carries most of the gain, but ViTA-Seg still leads C2F-Seg by 3.4 points of success rate and on both throughput metrics (SPF 3.95 vs. 3.41; OPF 4.11 vs. 3.40): more accurate object extent yields more reliable pick orderings. Second, at nearly identical frame counts (117 vs. 118), ViTA-Seg executes about 10% more grasp attempts in the same wall-clock session (631 vs. 576). Since the pipelines differ only in the perception module, that gap is inference overhead: half the parameters and no iterative decoding leaves more of a fixed shift available for motion rather than computation.

V. READINESS ASSESSMENT AND REMAINING GAPS

Against the workshop’s framing question, we read these results as follows. Occlusion-induced failure is real, separable, and worth 6–9 points of success rate plus a fifth of throughput in a representative suction cell; it is also the cheapest of the three failure causes to remove, since the fix is a ~ 100 M-parameter single-pass module inside the existing cycle rather than new sensing or new hardware. For an integrator, that is a favourable ratio.

TABLE II
REAL-WORLD BIN-PICKING PERFORMANCE OVER 5-HOUR UNATTENDED SESSIONS PER CONFIGURATION. RECOVERY COUNTS ARE A PROXY FOR OPERATOR-INTERVENTION LOAD.

Policy	Method	Total Grasp	Success Grasp	Total Frame	Success Rate (%)	SPF	OPF	Total Recovery
SSSP	CoAS-Net	1058	722	873	68.24	–	–	38
	CoAS-Net + ViTA-Seg	1050	783	912	74.57	–	–	26
SSMP	CoAS-Net	1293	824	312	63.72	2.64	2.59 ± 2.35	100
	CoAS-Net + ViTA-Seg	1588	1148	296	72.29	3.88	4.05 ± 3.13	67

TABLE III
MATCHED 2-HOUR SSMP SESSIONS DIFFERING ONLY IN THE AMODAL SEGMENTATION MODULE.

Method	Total Grasp	Success Grasp	Total Frame	Success Rate (%)	SPF	OPF	Total Recovery
CoAS-Net + C2F-Seg	576	402	118	69.79	3.41	3.40 ± 2.60	21
CoAS-Net + ViTA-Seg	631	462	117	73.21	3.95	4.11 ± 3.34	24

Four gaps remain. *Static-scene validity*: SSMP assumes the scene is unchanged during a planned sequence, yet lightweight or low-friction items can be displaced by a neighbouring grasp, silently invalidating the remaining ordering. Our object set does not stress this regime; fixing it likely means coupling amodal prediction with a world model reasoning about contact and friction. *Threshold sensitivity*: τ_{occ} and τ_{ov} are pixel-space constants tuned for one camera height and bin geometry, and would need re-derivation in metric units elsewhere. *Detector dependence*: the pipeline inherits every miss from upstream SAM 2. *Generalisation*: one object set, one gripper, one cell; benchmark transfer is suggestive but is not a second deployment.

VI. CONCLUSION

We measured what occlusion costs an autonomous bin-picking cell and what it costs to fix. Over 24 h of operation and 6,000+ real grasps, adding single-pass amodal segmentation and occlusion-hierarchy pick ordering improved success rate by 6.3 and 8.6 points under single- and multi-pick policies, raised throughput by over 22%, and cut recovery events by a third, with the accuracy and the computational cost of the perception module both converting measurably into operational output.

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